

ASSIGNMENT – 3

Index.html

```
<!DOCTYPE html>

<html>

<head>

  <title>Smart To-Do Manager</title>

  <script src="https://cdn.tailwindcss.com"></script>

</head>

<body class="bg-gray-100 min-h-screen p-4">

  <div class="max-w-3xl mx-auto bg-white p-6 rounded shadow">

    <h1 class="text-3xl font-bold mb-4 text-center">Smart To-Do Manager</h1>

    <!-- Add Task -->

    <div class="grid md:grid-cols-4 gap-2 mb-4">

      <input id="title" type="text" placeholder="Task title" class="border p-2 rounded">

      <input id="category" type="text" placeholder="Category (Work/Personal)" class="border p-2 rounded">

      <input id="dueDate" type="date" class="border p-2 rounded">

      <button onclick="addTask()" class="bg-blue-500 text-white rounded p-2">Add</button>

    </div>

    <!-- Search & Filter -->

    <div class="grid md:grid-cols-3 gap-2 mb-4">

      <input id="search" oninput="renderTasks()" placeholder="Search..." class="border p-2 rounded">

      <select id="statusFilter" onchange="renderTasks()" class="border p-2 rounded">

        <option value="">All Status</option>
```

```

    <option value="completed">Completed</option>
    <option value="pending">Pending</option>
  </select>

  <input id="categoryFilter" oninput="renderTasks()" placeholder="Filter by Category"
  class="border p-2 rounded">
</div>

<!-- Task List -->
<div id="taskList" class="space-y-2"></div>

</div>

<script>
const API = "http://localhost:5000/tasks";
let tasks = [];

async function fetchTasks() {
  const res = await fetch(API);
  tasks = await res.json();
  renderTasks();
}

async function addTask() {
  const title = document.getElementById("title").value;
  const category = document.getElementById("category").value;
  const dueDate = document.getElementById("dueDate").value;

  if (!title) return alert("Task title required");

  await fetch(API, {
    method: "POST",

```

```
headers: { "Content-Type": "application/json" },
body: JSON.stringify({ title, category, dueDate })
});
```

```
document.getElementById("title").value = "";
fetchTasks();
}
```

```
async function toggleTask(id) {
  await fetch(`${API}/${id}`, { method: "PUT" });
  fetchTasks();
}
```

```
async function deleteTask(id) {
  await fetch(`${API}/${id}`, { method: "DELETE" });
  fetchTasks();
}
```

```
function renderTasks() {
  const search = document.getElementById("search").value.toLowerCase();
  const statusFilter = document.getElementById("statusFilter").value;
  const categoryFilter = document.getElementById("categoryFilter").value.toLowerCase();

  const filtered = tasks.filter(task => {
    const matchesSearch = task.title.toLowerCase().includes(search);
    const matchesCategory = task.category?.toLowerCase().includes(categoryFilter);
    const matchesStatus =
      statusFilter === "" ||
      (statusFilter === "completed" && task.completed) ||
      (statusFilter === "pending" && !task.completed);
```

```
    return matchesSearch && matchesCategory && matchesStatus;
  });
```

```
const taskList = document.getElementById("taskList");
taskList.innerHTML = "";
```

```
filtered.forEach(task => {
  taskList.innerHTML += `
    <div class="flex justify-between items-center p-3 border rounded">
      <div>
        <p class="${task.completed ? 'line-through text-gray-500' : ''}">
          ${task.title}
        </p>
        <small class="text-sm text-gray-500">
          ${task.category || "No Category"} | Due: ${task.dueDate || "N/A"}
        </small>
      </div>
      <div class="space-x-2">
        <button onclick="toggleTask(${task.id})"
          class="px-3 py-1 bg-green-500 text-white rounded">
          ✓
        </button>
        <button onclick="deleteTask(${task.id})"
          class="px-3 py-1 bg-red-500 text-white rounded">
          🗑️
        </button>
      </div>
    </div>
  `;
});
```

```
});  
}  
  
fetchTasks();  
</script>  
  
</body>  
</html>
```

Backend/server.js

```
const express = require("express");  
const fs = require("fs");  
const cors = require("cors");  
const app = express();  
  
app.use(cors());  
app.use(express.json());  
  
const DB_FILE = "./db.json";  
  
// Helper function to read DB  
const readData = () => {  
  const data = fs.readFileSync(DB_FILE);  
  return JSON.parse(data);  
};  
  
// Helper function to write DB  
const writeData = (data) => {  
  fs.writeFileSync(DB_FILE, JSON.stringify(data, null, 2));  
};
```

```
// GET all tasks

app.get("/tasks", (req, res) => {
  const data = readData();
  res.json(data.tasks);
});

// POST new task

app.post("/tasks", (req, res) => {
  const data = readData();
  const newTask = {
    id: Date.now(),
    title: req.body.title,
    category: req.body.category,
    dueDate: req.body.dueDate,
    completed: false
  };

  data.tasks.push(newTask);
  writeData(data);
  res.status(201).json(newTask);
});

// PUT toggle status

app.put("/tasks/:id", (req, res) => {
  const data = readData();
  const task = data.tasks.find(t => t.id === req.params.id);

  if (!task) return res.status(404).json({ message: "Task not found" });
```

```
task.completed = !task.completed;
writeData(data);
res.json(task);
});

// DELETE task
app.delete("/tasks/:id", (req, res) => {
  const data = readData();
  data.tasks = data.tasks.filter(t => t.id !== req.params.id);
  writeData(data);
  res.json({ message: "Deleted successfully" });
});

app.listen(5000, () => {
  console.log("Server running on http://localhost:5000");
});
```

Backend/db.json

```
{
  "tasks": []
}
```

Backend/package.json

```
{
  "name": "smart-todo-backend",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
}
```

```
"dependencies": {  
  "cors": "^2.8.5",  
  "express": "^4.18.2"  
}  
}
```


Output page

Smart To-Do Manager





home-work
work | Due: 2004-11-02

 

home work
work | Due: 2004-11-02

 

test task
work | Due: 2004-03-02

 

exercise
personal | Due: 2025-10-06

 

Stored database

```
1 {
2   "tasks": [
3     {
4       "id": 1771937134556,
5       "title": "home work",
6       "category": "work",
7       "dueDate": "2004-11-02",
8       "completed": true
9     },
10    {
11      "id": 1771937154803,
12      "title": "home work",
13      "category": "work",
14      "dueDate": "2004-11-02",
15      "completed": false
16    },
17    {
18      "id": 1771937171780,
19      "title": "test task",
20      "category": "work",
21      "dueDate": "2004-03-02",
22      "completed": false
23    }
24  ]
25 }
```